Airport Security Check-Point

Presenters :
Dennis Michael T (231BCADA13)
Anthony Christopher P (231BCADA22)

Introduction

•Air transportation is one of the most commonly used modes of travel worldwide because it is fast, convenient and provides global connectivity.

•The number of air passengers is increasing every year, which makes airport safety and security a major concern.

•Among all airport operations, the airport security checkpoint plays a very important role.

•It helps in protecting passengers, crew members, aircraft and airport infrastructure.

•Security checkpoints act as the first line of defense against illegal activities.

•These activities include terrorism, smuggling of prohibited items and other potential security threats.

Problem Statement

1. Increasing air passenger traffic puts high pressure on airport security checkpoints.

2. Threat detection depends on multiple factors like AI risk scoring, passenger behavior, and staff experience.

3. Balancing security accuracy with operational efficiency is a major challenge.

4. A simulation-based model is needed to analyze and optimize airport checkpoint performance

Related works

[1] Pik et al.
They discussed recent advancements in airport security technologies, especially AI-based risk prediction systems.
Their study highlights improved detection accuracy and speed, while also raising ethical and privacy concerns.
[2] Schwaninger et al. (EDSCB Study)
They examined the implementation of Explosive Detection Systems for cabin baggage screening.
The study compared partial automation (human-assisted) and fully automated machine decision systems.
[3] Uchronski et al.
They proposed a fuzzy inference system to evaluate prohibited item detection effectiveness.
Results showed that upgrading equipment and increasing staff training improves system performance.
[4] Schwaninger et al. (2D vs 3D Imaging)
They analyzed the transition from 2D to 3D imaging systems in airport screening.
The study emphasizes the need for proper screener training and use of on-screen alarm resolution protocols (OSARP).
[5] Zhou et al.
They studied how different checkpoint network structures affect passenger flow and system efficiency.
Open network structures and better staff allocation significantly improve operational performance.

About the Dataset

Synthetic Airport Security Checkpoint Dataset

1. Simulates real-world passenger screening operations

2. Includes demographic, behavioral, AI-based and operational factors

3. Designed to analyze factors influencing security threat detection

4. Target Variable: **Security_Threat_Level (0 = Safe, 1 = Threat)**

5. Supports binary classification modeling for intelligent decision-making

Objective:
To study how operational and behavioral variables contribute to identifying potential security threats.

Key Variables & Influence on Threat Detection
Strong Direct Influence

1. Prohibited_Item_Detected
2. Baggage_Screening_Result

Moderate Influence

1. AI_Risk_Score
2. Passenger_Preparedness
3. Security_Staff_Experience

Indirect Influence

1. Passenger_Age
2. Screening_Time
3. Passenger_Throughput_Level

The dataset captures both technical and human factors affecting threat classification.

Importance & Research Value

This dataset is suitable for:

1. Security risk prediction modeling
2. Screening efficiency analysis
3. Evaluating staff experience impact
4. Studying operational pressure effects
5. AI-assisted decision-support system research

Conclusion:
The dataset provides a comprehensive synthetic model of airport checkpoint operations and supports predictive modeling, simulation studies, and checkpoint optimization research.

Passenger_	Passenger_Preparedness	Baggage_Screening_Result	Prohibited_Item_Detected	AI_Risk_Score	Security_Staff_Experience	Screening_Time	Passenger_Throughput_Level	Security_Threat_Level
56	0	0	0	19	22	1.87		0
69	0	0	0	21	15	2.7	1	0
46	0	1	0	60	13	4.49	1	1
32	0	0	0	32	23	3.07	0	0
60	0	0	0	21	19	1.99	2	0
25	1	0	0	31	9	1.98	1	0
78	1	1	0	93	4	6.09	0	1
	0	0	0	28	21	3.31	2	0
56	1	0	0	18	8	3.06		0

Methodology – Approach Used

Main Approach: Agent-Based Modeling (ABM)

•Agents act independently based on defined rules
•System behavior emerges from interactions over time

In Our Simulation:

•Each passenger acts independently
•Each passenger has a threat probability
•Security attempts to detect threats
•Overall airport threat level evolves over time

Methodology Type:
Data-driven Agent-Based Simulation using probabilistic threat modeling

Techniques Combined:
•Dataset preprocessing
•Risk-based probability modeling
•Stochastic (random) simulation
•Time-step iteration

Agents (Decision-Making Entities)

1. Passenger Agent

•Arrives at airport

•Has probability of being a threat

Formula:

$$threat_probability = 0.05 + 0.3 \times normalized_risk$$

Behavior:

•Probabilistic decision-making

•Influenced by AI Risk Score

•Independent from other passengers

2. Security Agent (AI Detection Model)

•Checks each potential threat

•Detects threats with fixed probability

$$detection_probability = 0.75$$

•Simulates AI-based detection system

•Detection is not 100% perfect

Environment (System Context)

1. Time
- Each iteration = one time step
- `simulation_time = len(df)`
- System evolves step-by-step

2. AI Risk Score
- Normalized (0–1)
- Higher score → higher threat probability

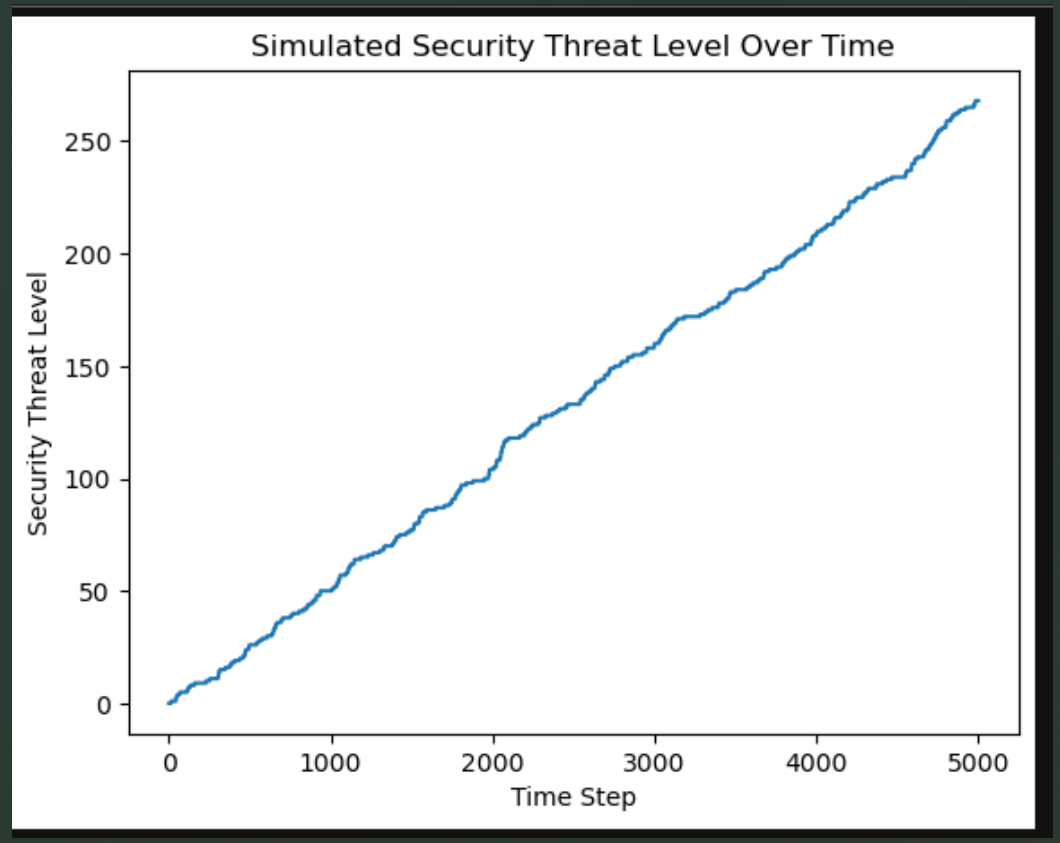
3. Passenger Throughput Level
- Controls number of passengers per step
- Higher throughput → more threats
- Creates operational pressure

4. Current Threat Level (State Variable)
`current_threat_level += (threats - detected)`

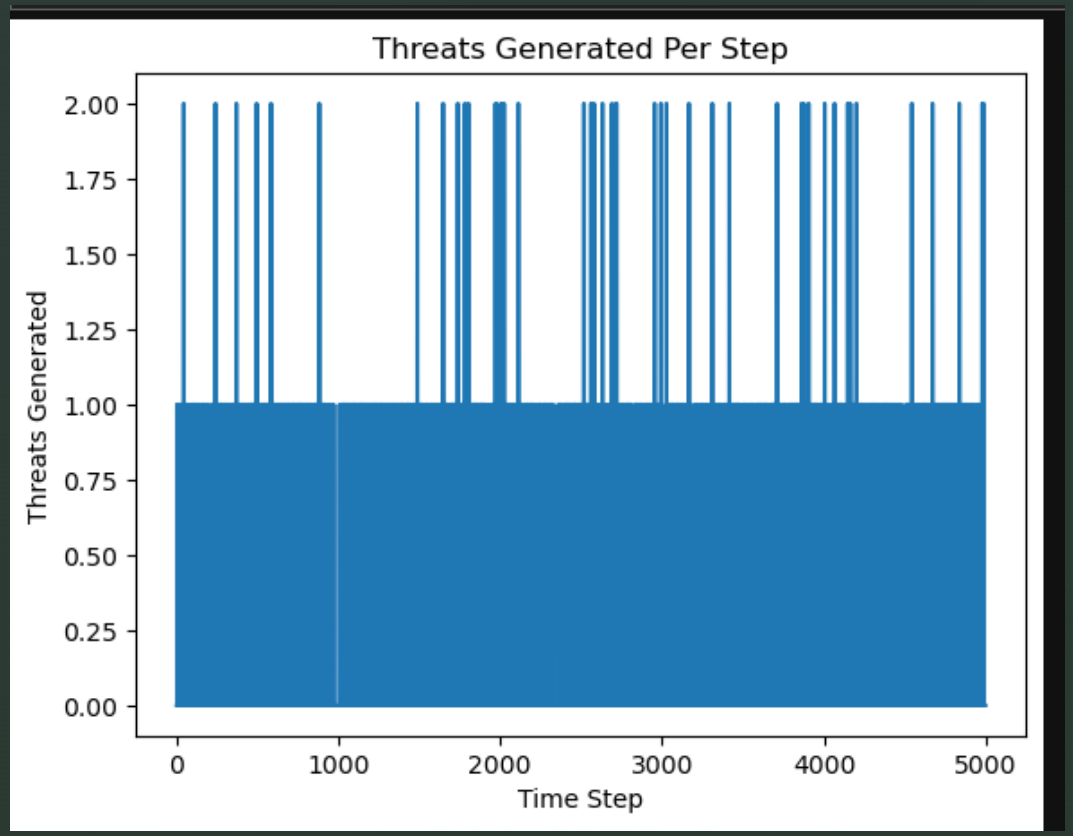
Represents:
- Unresolved threats
- System risk level
- Overall security performance

If detection is weak → threat level increases
If detection is strong → threat level stabilizes

Threat Level Over Time (Line Graph)



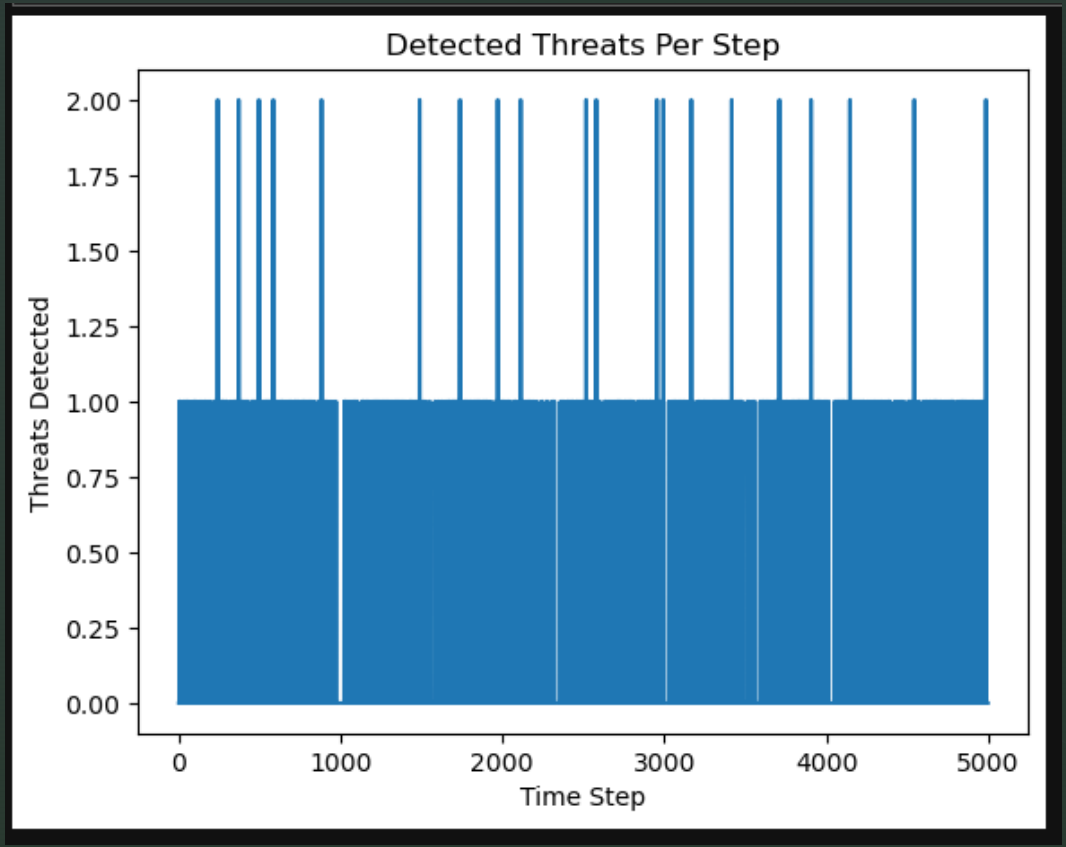
1. This graph shows how the overall airport security threat level changes at each time step in the simulation.
2. If the line increases, it means more threats were generated than detected at that time.
3. If the line remains stable or decreases, it shows that the security system is effectively detecting threats.
4. This represents the target variable of our simulation



Threats Generated Per Step (Line Graph)

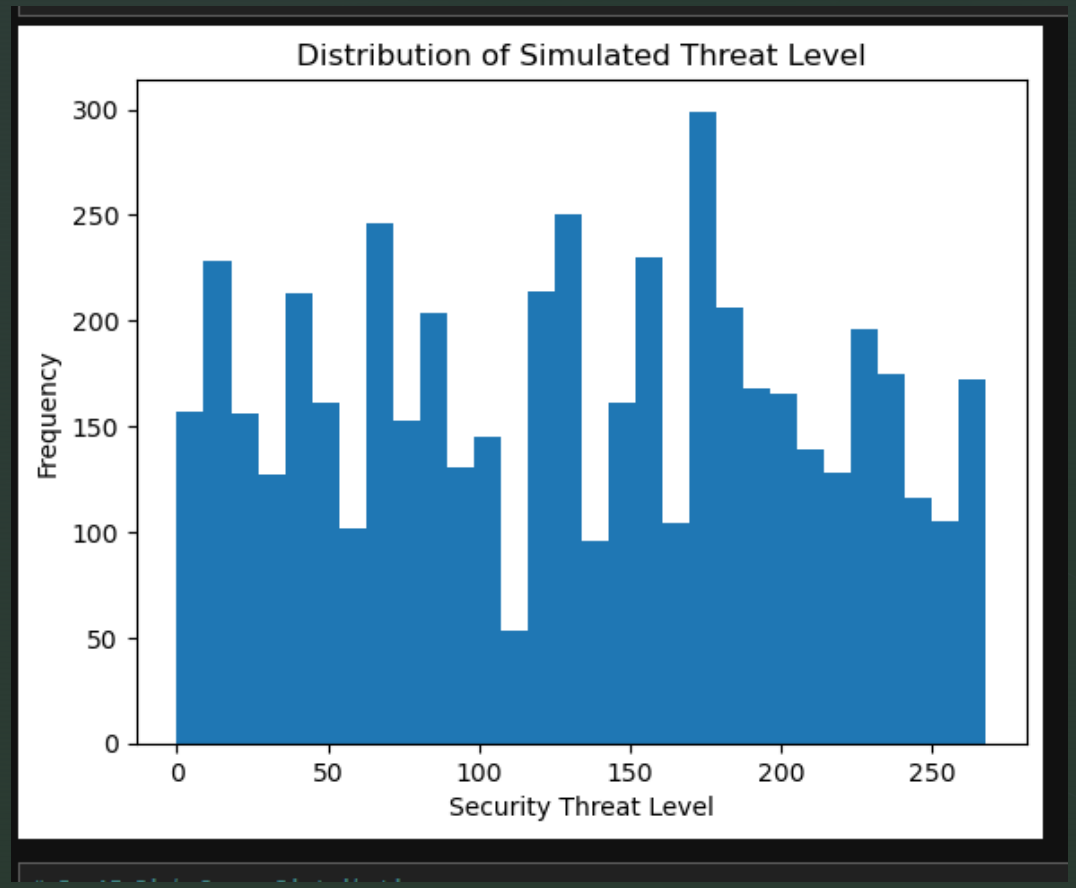
1. This graph shows how many new threats were created at each time step.
2. The number of threats depends on passenger count and AI risk score.
3. Higher spikes indicate higher passenger traffic or higher AI risk.
4. This shows the risk intensity in the environment.

Detected Threats Per Step (Line Graph)



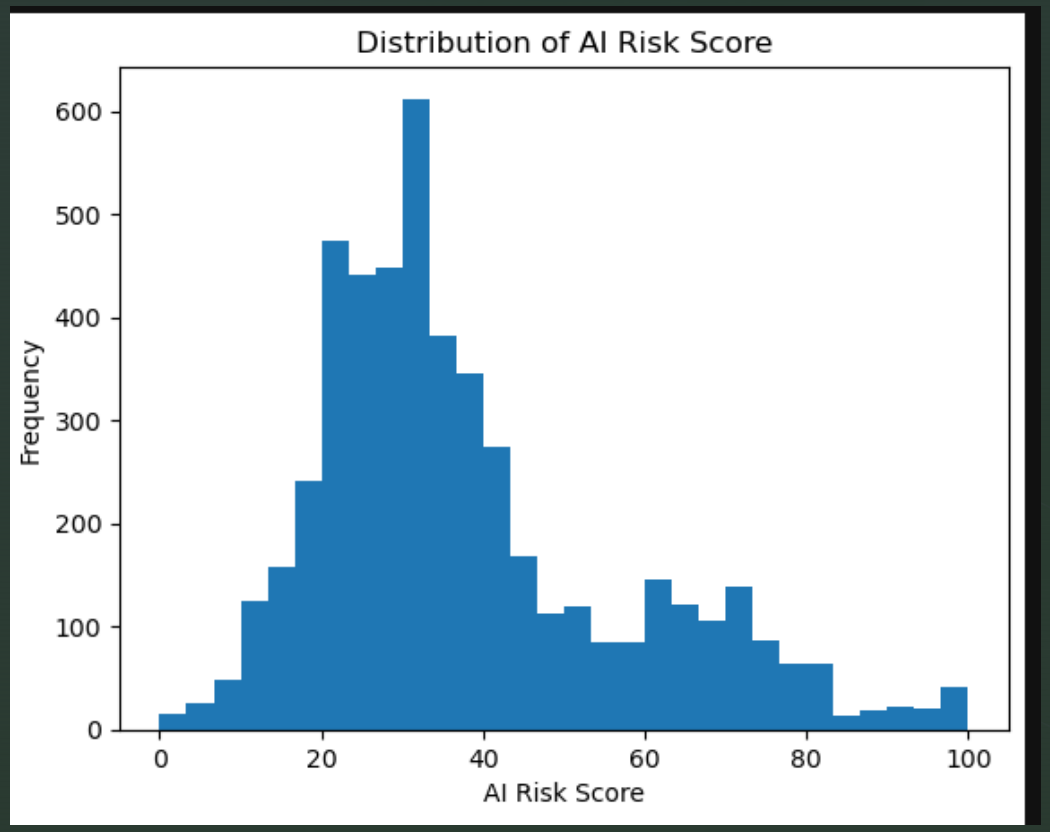
1. graph shows how many threats were successfully detected by the security system.
2. Detection is based on the 75% YOLO-style detection probability.
3. If this line is close to the generated threats graph, the system is performing well.
4. This measures security system performance

Distribution of Threat Level (Histogram)

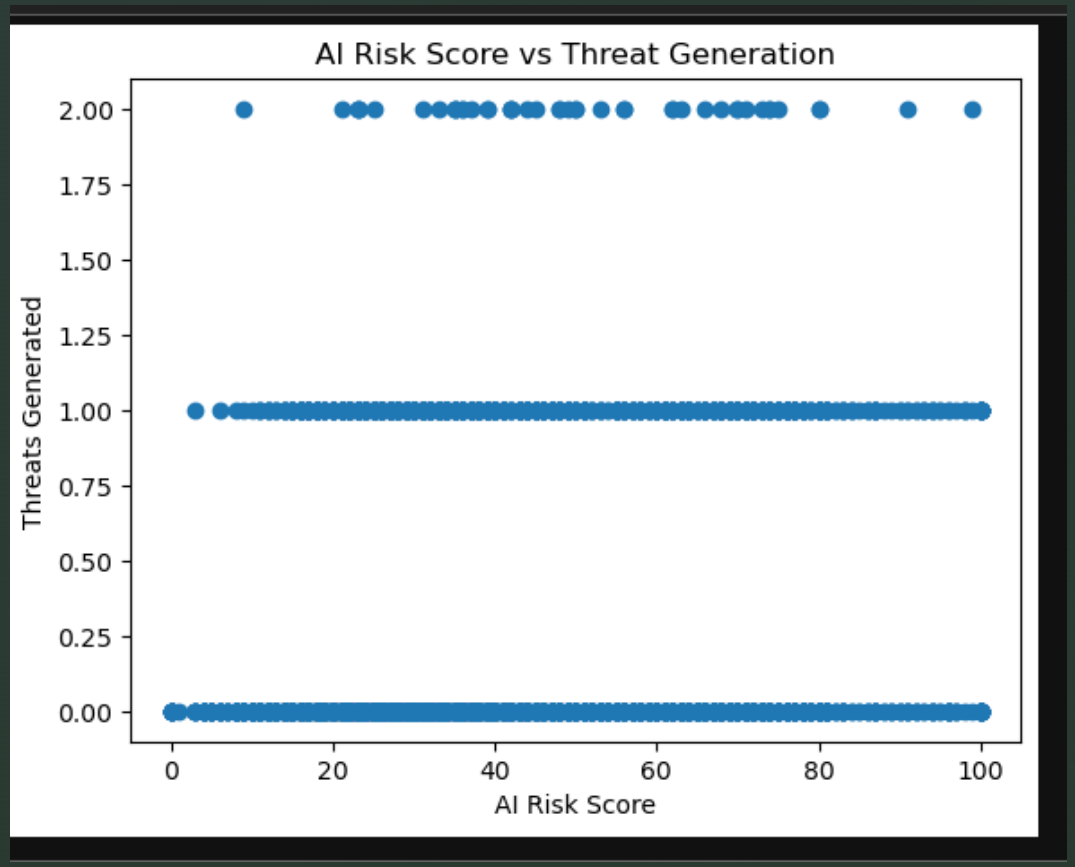


1. This graph shows how frequently different threat levels occurred during the simulation.
2. If most values are near zero, security is under control.
3. If higher threat levels appear frequently, it indicates system weakness.
4. This shows overall security stability.

AI Risk Score Distribution (Histogram)

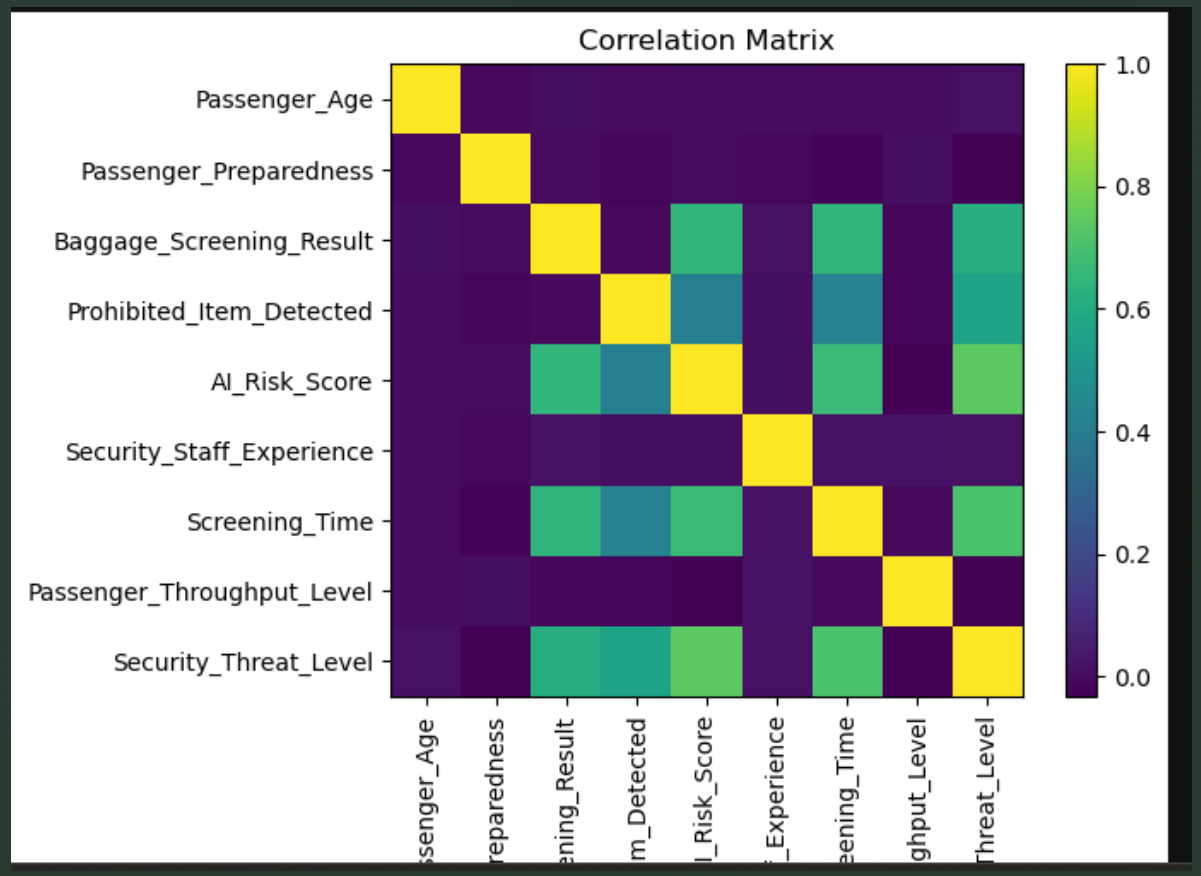


1. This graph shows how AI risk scores are distributed in the dataset.
2. A uniform distribution means risk is spread evenly.
3. A right-skewed distribution indicates many high-risk situations.
4. This helps understand environmental risk patterns



AI Risk Score vs Threat Generation (Scatter Plot)

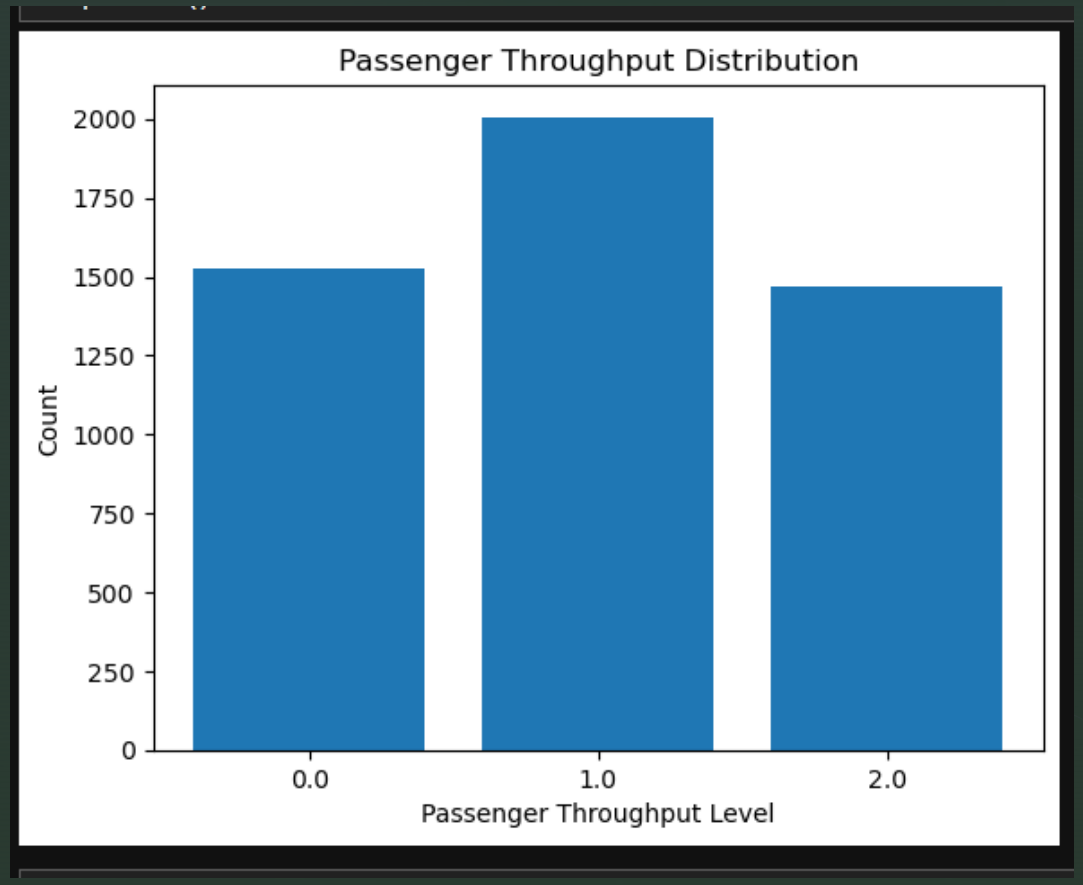
1. Each point represents AI risk score vs threats generated at that time.
2. If points form an upward trend, higher AI risk leads to more threats.
3. This validates that AI risk score influences threat probability.
4. This proves the model relationship.



Correlation Matrix (Heatmap)

1. This graph shows relationships between all numeric variables.
2. Values close to +1 indicate strong positive correlation.
3. Values near 0 indicate weak or no relationship.
4. This helps in feature importance analysis.

Passenger Throughput Distribution (Bar Chart)



1. This graph shows how passenger levels are distributed.
2. Higher throughput means more agents entering the system.
3. More passengers increase the probability of threat generation.
4. This represents environment pressure on security.

Results

1. **Classification Model Performance**

The Random Forest model successfully classified passengers into:
- **Safe (0)**
- **Threat (1)**

Key Observations:

- **Strong Positive Influence:**
  - Prohibited_Item_Detected
  - Baggage_Screening_Result
  - AI_Risk_Score
- **Negative Influence:**
  - Passenger_Preparedness
  - Security_Staff_Experience

The model showed clear separation between high-risk and low-risk passengers, indicating that AI-based risk scoring and screening results are strong predictors of security threats.

2. Simulation Results (Agent-Based Model)

- **Threat Level Behavior**

The threat level changed dynamically over time depending on detection performance. When detection was efficient, the system remained stable. When detection lagged behind generated threats, unresolved threats accumulated.

- **Impact of AI Risk Score**

Higher AI Risk Scores led to higher threat probabilities.
This confirms that the probabilistic modeling approach is functioning correctly.

- **Impact of Passenger Preparedness**

Higher passenger preparedness reduced operational delays and lowered predicted threat probability.

- **Impact of Staff Experience**

More experienced security staff improved detection consistency and reduced false classifications.

- **Throughput Effects**

High Passenger_Throughput_Level increased operational pressure, which sometimes affected detection balance and threat stabilization.

Discussion

The findings demonstrate that airport security threat detection is **multi-factor dependent**. It is not controlled by a single variable but by a combination of:

- AI-based risk scoring
- Screening results
- Passenger behavior
- Staff expertise
- Operational pressure

The integration of Agent-Based Modeling allowed dynamic analysis of how these factors interact over time. Unlike static machine learning models, the simulation captured system evolution and unresolved threat accumulation.

Conclusion

- **Key Findings**

•AI-based risk scoring and baggage screening results strongly predict threat classification.
•High passenger throughput (operational pressure) impacts screening efficiency.
•Logistic Regression provides an interpretable and effective threat prediction model.

- **Dataset Significance**

The dataset is suitable for:
•Security risk prediction modeling
•Operational efficiency analysis
•AI-assisted decision-support research
•Simulation-based checkpoint optimization

- **Final Insight**

Integrating AI systems with trained personnel and improving passenger preparedness can significantly enhance airport security performance, while maintaining efficiency and addressing ethical concerns ensures long-term system reliability.